



INTRODUCTION TO ROBOTICS

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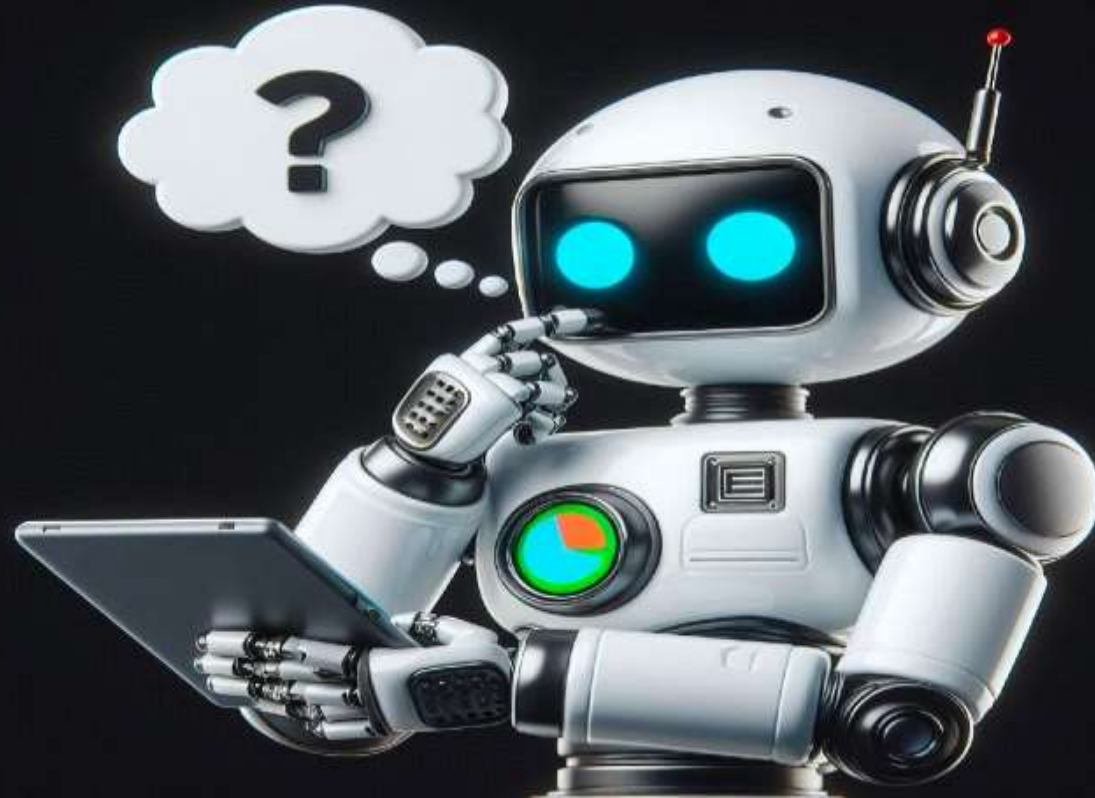
Agenda of the Workshop

Key focusing Areas:

- . Introduction to Robotics and IoT technology
- . Requirement for Robots technology
- . Robotics/ IoT Applications
- . Robotics Career Guidance
- . Hands-on Project for Robotics (Practical Session)
 - Online tools for IoT/ Robot project
 - Make a simple robot system



What is a Robot...?





What is a Robot...?

smart machines that can work by themselves or with instructions.

it combines:

- Electronics (circuits, sensors, motors)
- Programming (to control behavior)
- Mechanical parts (structure and movement)

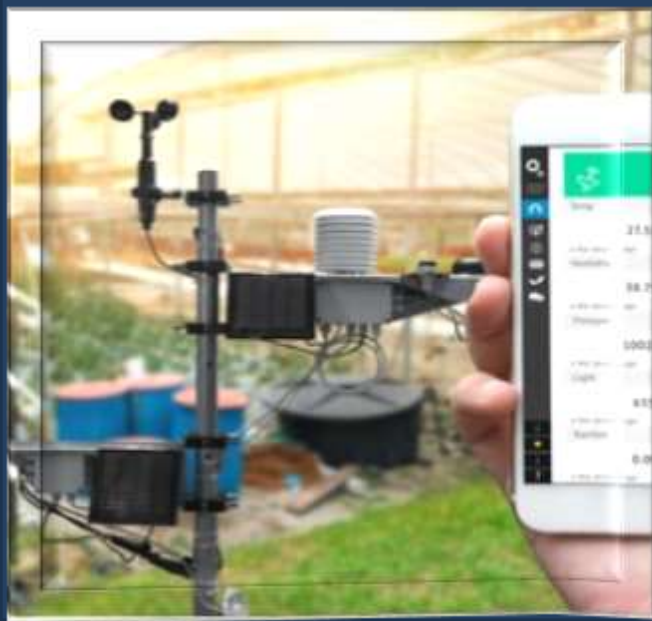


What is a IoT ...?



What is a IoT ...?

- IoT refers to a network of interconnected devices that communicate and exchange data over the internet.
- These devices include sensors for data collection, communication modules for data transfer, and actuators for control and automation.



ROBOTICS TECHNOLOGY

Most industrial robots have at least the following five parts:



Sensors



Actuators



Controllers



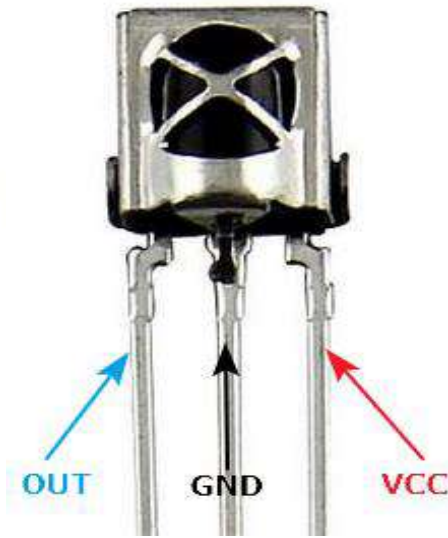
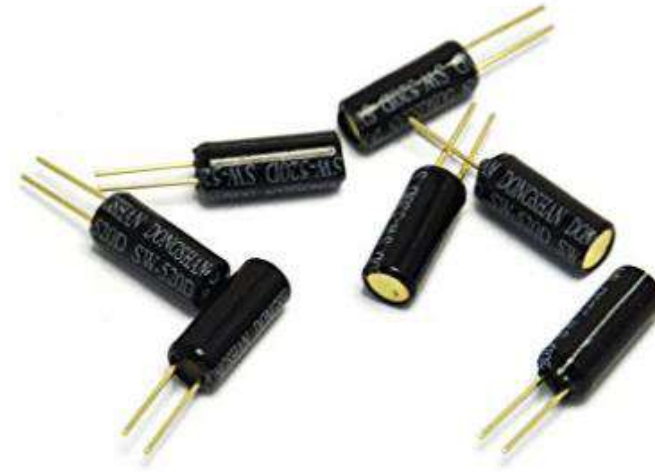
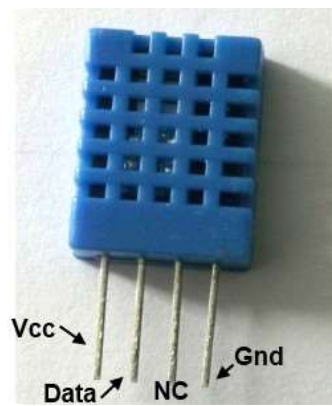
Arms



Sensors

Used for collecting data from the physical environment.

- PIR Sensors
- IR Sensors
- Ultrasonic Sensors
- Accelerometers
- Magnet Sensors
- Gas Sensors
- Temp Sensors
- Humidity Sensors



Actuators

These are used to perform actions based on received instructions.

- LED
- LCD Display
- Motors
- Buzzer
- Relay
- Valve



Controllers

- PIC microcontroller
- Atmel microcontroller
- Intel microcontroller
- EPSON microcontroller
- PLC
- Raspberry Pi

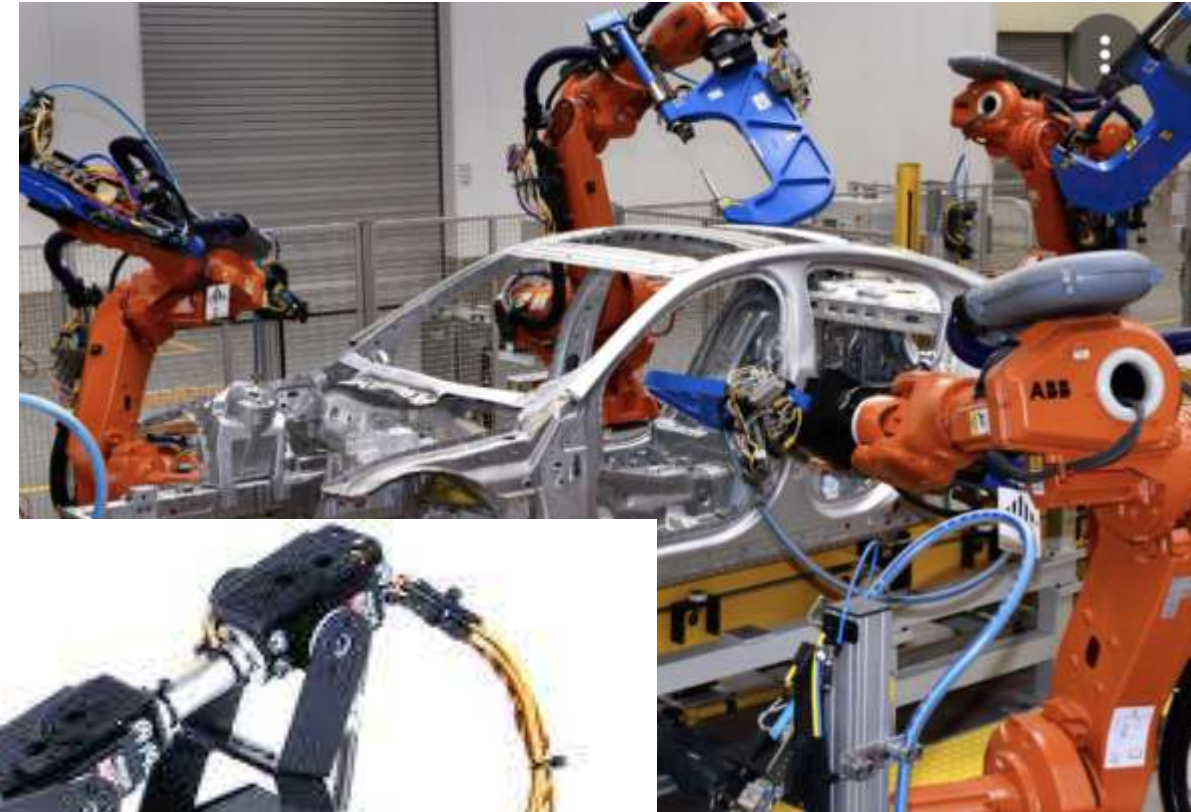


Controllers (such as microcontrollers or processors) are a critical component of Robotics/ IoT devices, as they function as the "brain" of the system.



Robot Arms/ Hardware

Robot Arms are mechanical devices designed to simulate the movements and functionalities of a human arm.



Robotics Applications

- Assembly lines robots
- Welding robots
- Painting robots
- Healthcare Surgical Robots
- Robotic vacuum cleaners



IoT Applications

- Smart Homes
- Smart City
- Self-driven Cars
- Wearable health trackers
- IoT Retail Shops
- Agriculture
- Safety and Security



Robotics Career Guidance

Current Trends and Opportunities in Sri Lanka

- **Automation in Manufacturing:**

Companies in Sri Lanka are accepting robotic automation to improve efficiency in production lines.

- **Agricultural Robotics:**

Development of robotics solutions for care farming, such as automated harvesting and crop monitoring.

- **Healthcare Robotics:**

Emerging interest in robotic-assisted surgery and rehabilitation devices in the healthcare sector.

- **Robotics Research and Education:**

Universities and institutions are focusing on robotics education and projects, offering more courses and competitions to nurture talent.

Robotics Career Guidance

Current Trends and Opportunities in other country

- **Industrial Robotics:** ([example](#))

Japan is a global leader in robotics, with companies like **FANUC**, **Yaskawa**, and **Kawasaki** producing advanced industrial robots.

- **Service Robots:** ([example](#))

Innovations in hospitality, retail, and elder care robots, such as Pepper and Paro.

- **Healthcare Robots:** ([example](#))

Extensive use of robots for elder care, surgical assistance, and rehabilitation.

- **AI Integration:**

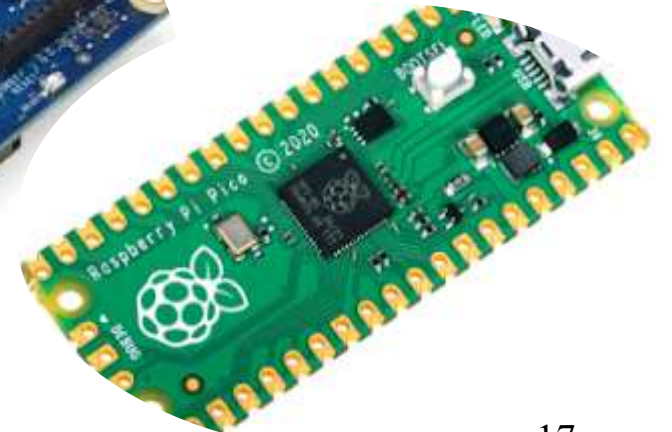
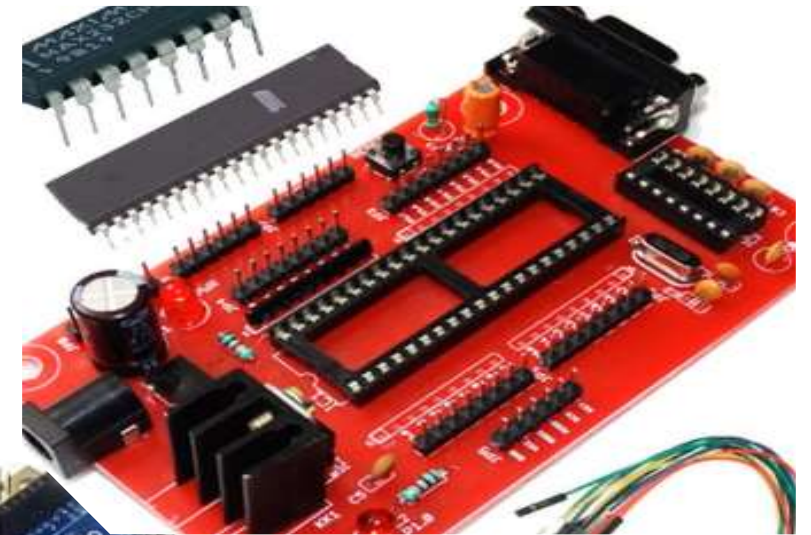
Robotics combined with artificial intelligence for autonomous decision-making in various industries.

Hands-on Project for Robotics (Practical Session)

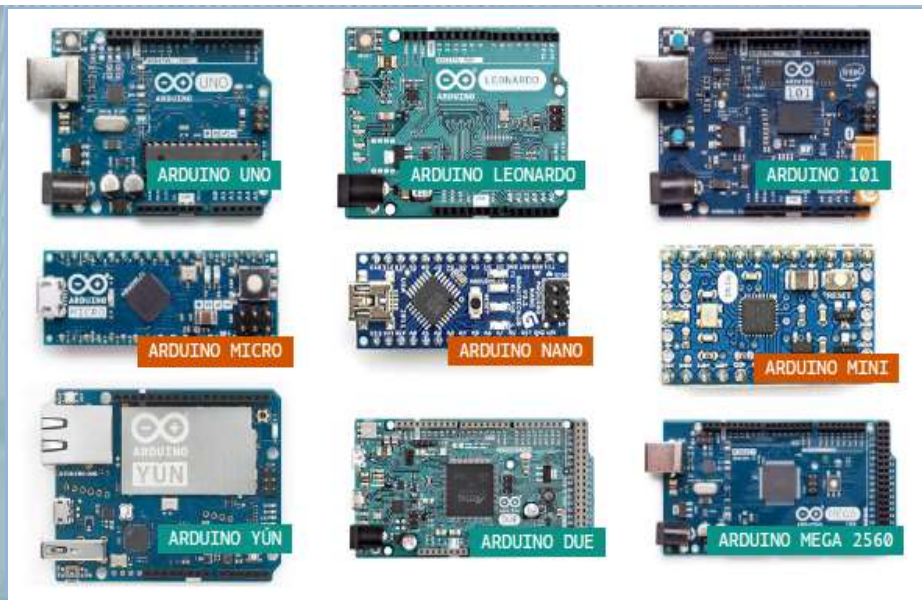
Microcontroller Development Boards

- A **microcontroller development board** is a circuit board that can be used to **program** and **test** with a specific microcontroller board's capabilities.

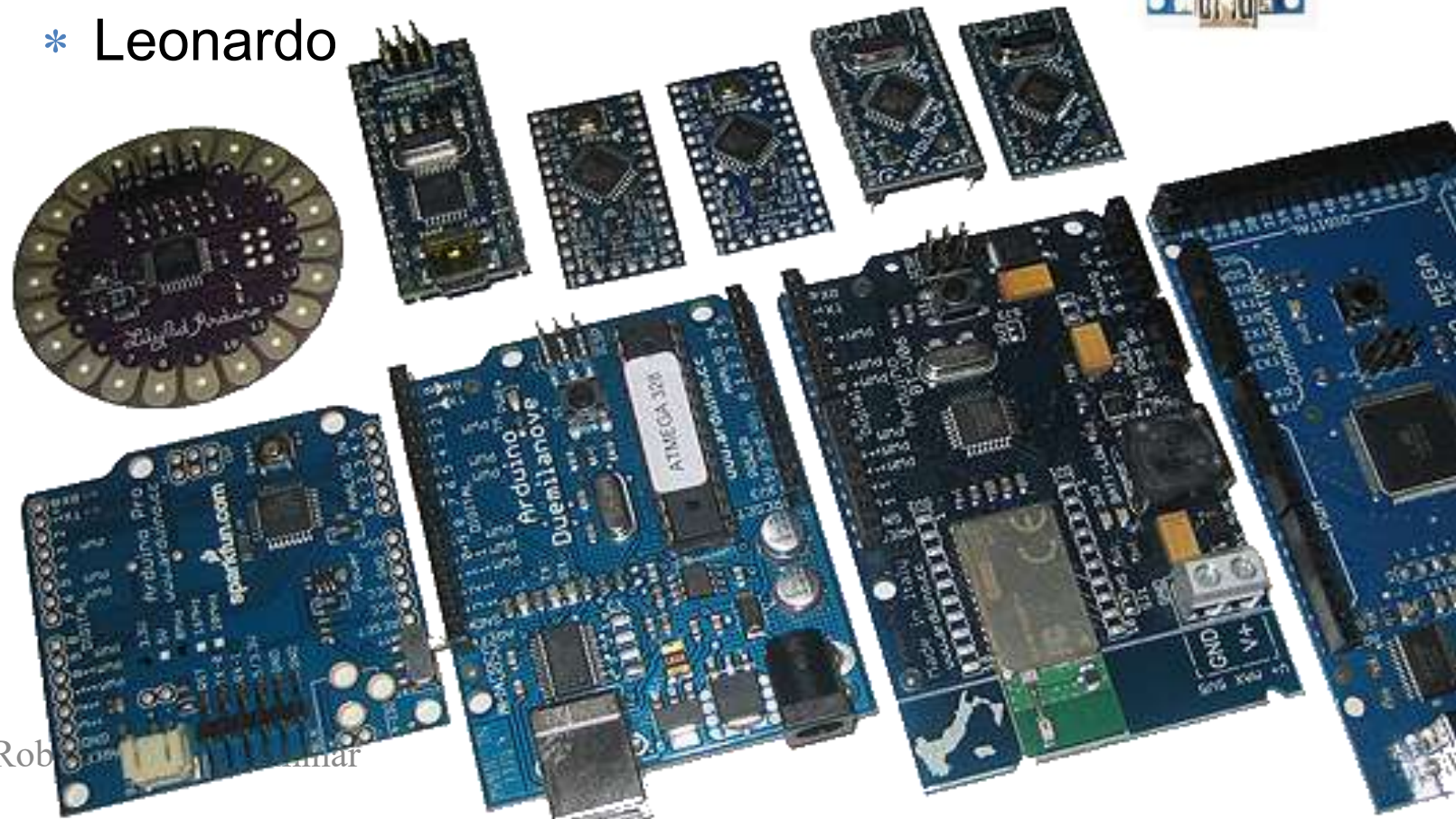
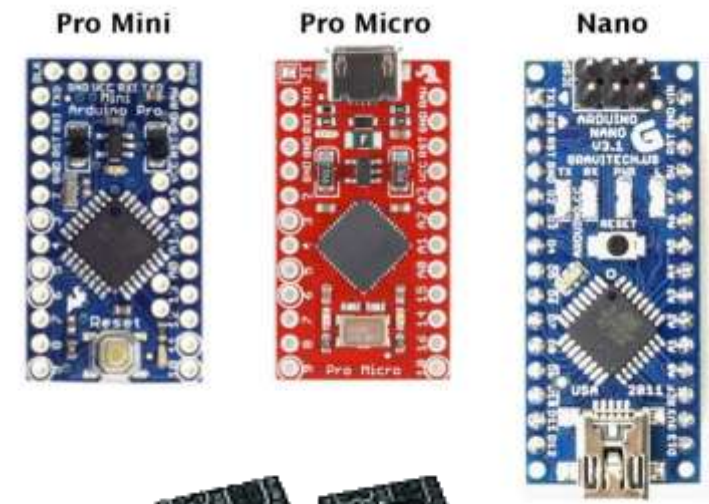
- **Arduino Board**
- **Raspberry Pi Board**
- **ESP8266 Board**
- **PIC Board**
- **PLC Board**



Arduino Boards



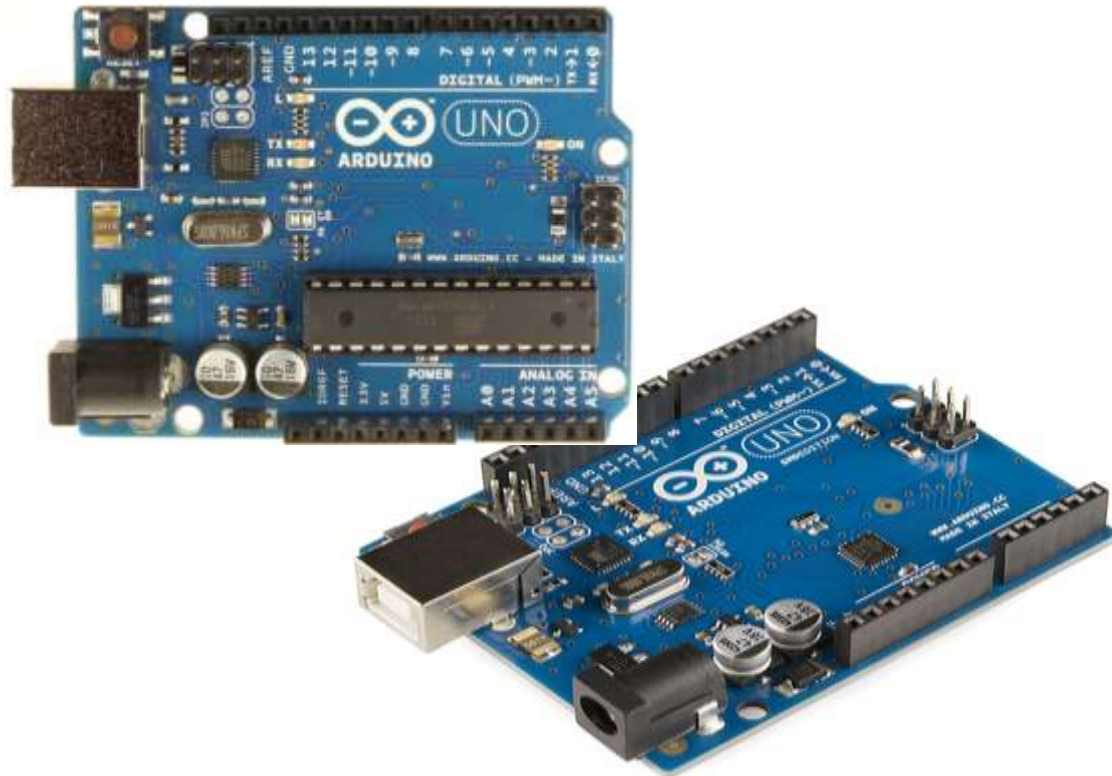
- * Due
- * Mega
- * Uno
- * Nano
- * Mini
- * Micro
- * Leonardo



Arduino Uno Board Vs Mega Board

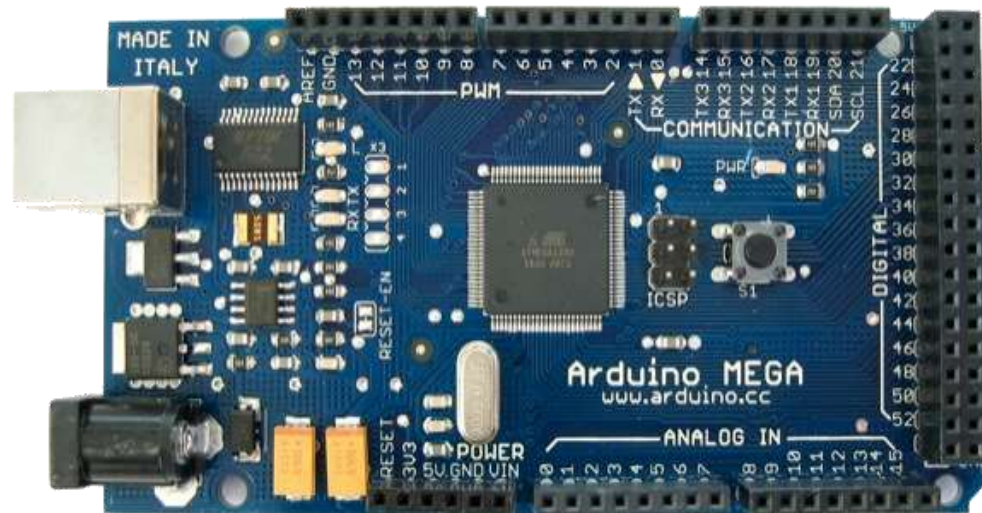
UNO – Atmega 328P chip

- Digital pins -14
- Memory - 32 kb
- Ram- 2 kb



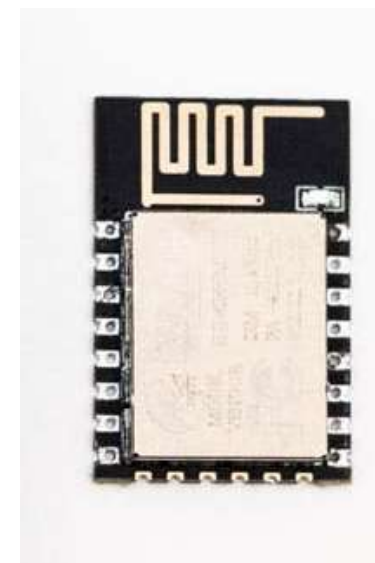
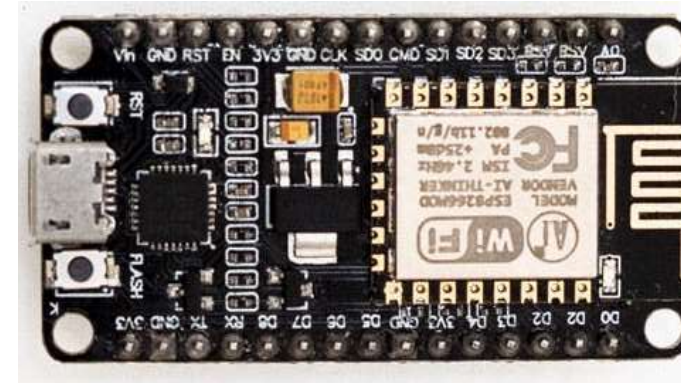
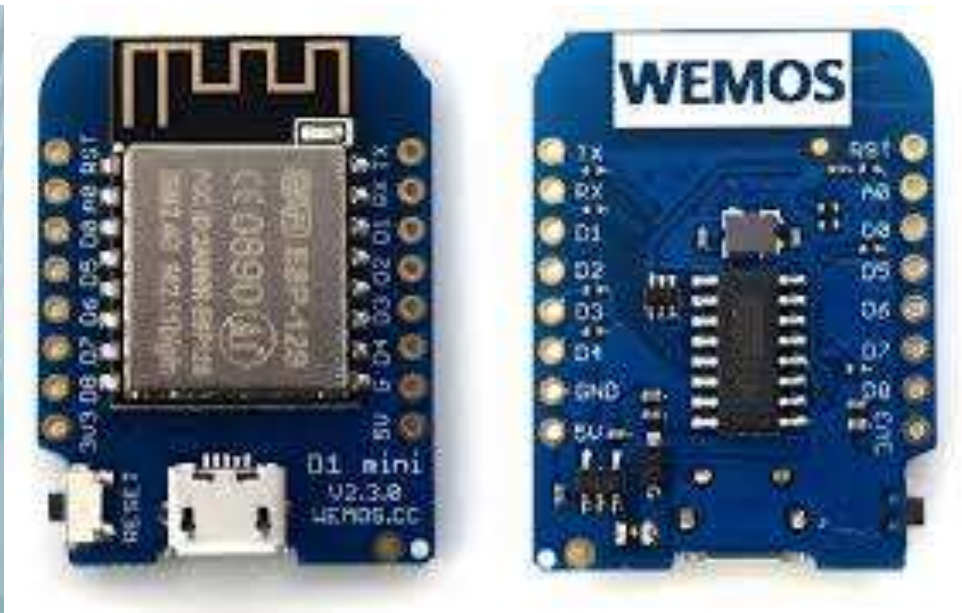
Mega – Atmega 2560 chip

- Digital pins -54
- Memory -256 kb
- Ram- 8kb



Select suitable Boards for Robotics & IoT

- * Wemos D1
- * NodeMCU
- * ESP32 Boards
- * Nano 33 IoT
- * MKR WiFi 1010

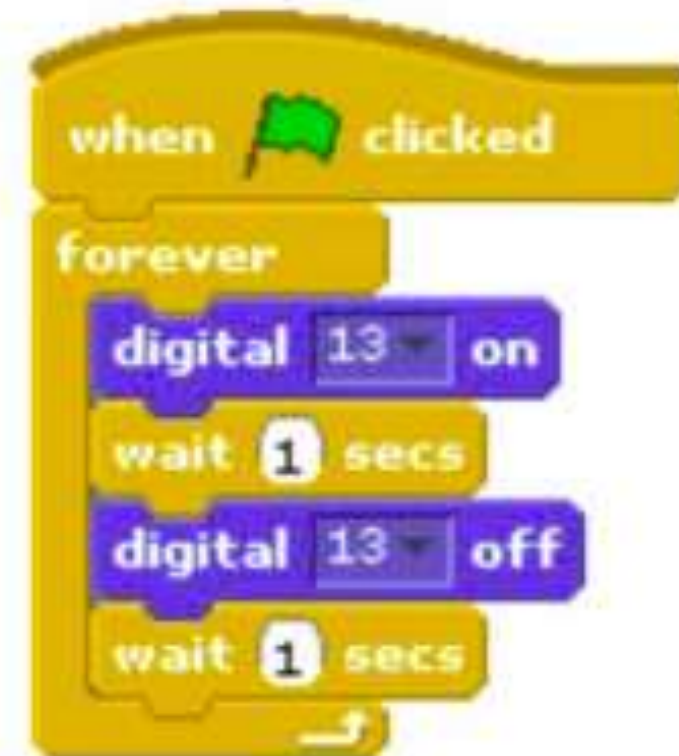


Write control Programme - Cording Vs Scratch

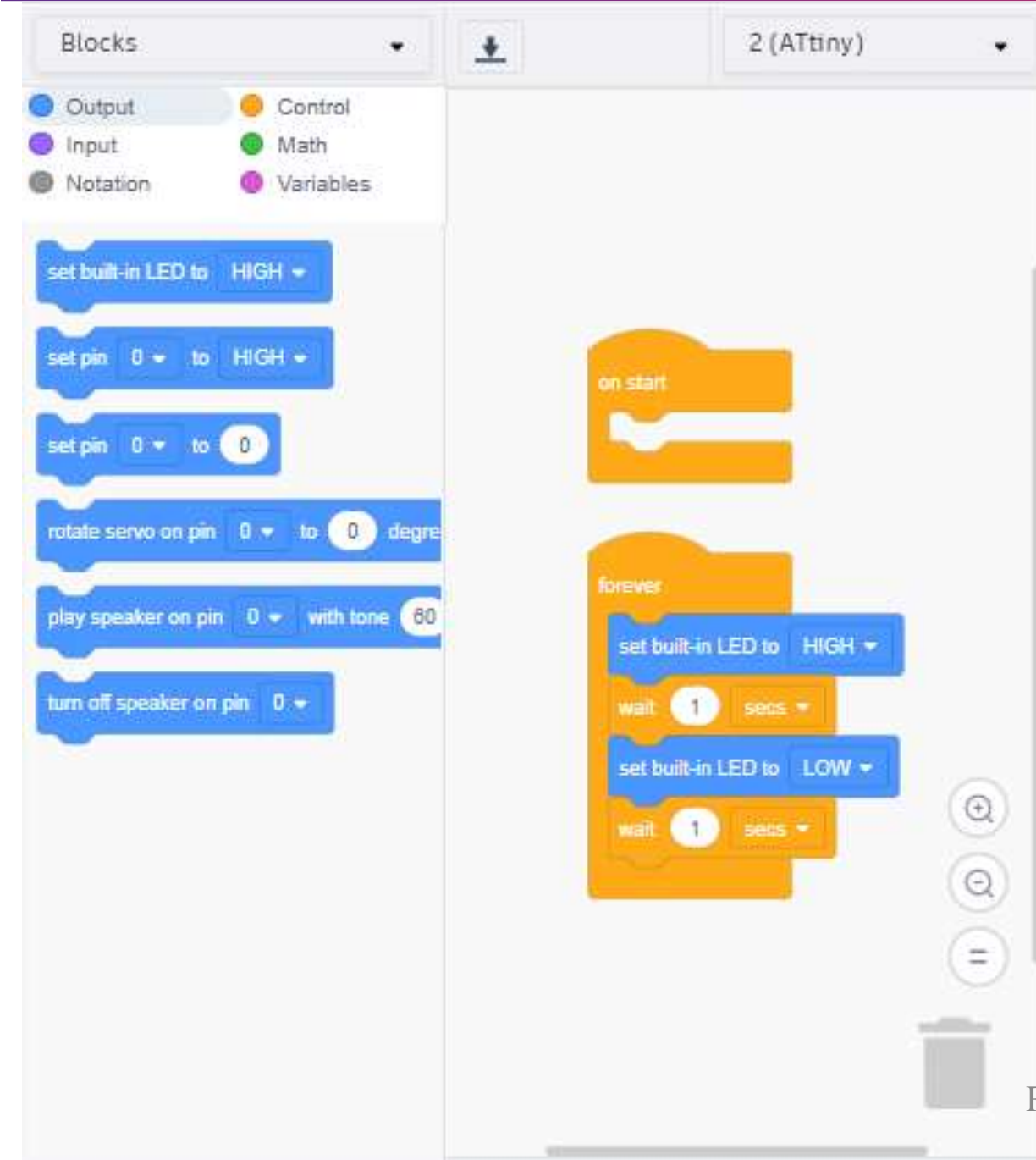
Arduino

```
void setup() {  
  pinMode(13, OUTPUT);  
}  
  
void loop() {  
  digitalWrite(13, HIGH);  
  delay(1000);  
  digitalWrite(13, LOW);  
  delay(1000);  
}
```

S4A



Write control Programme – Scratch



Scratch for Arduino

- Use block to write programme

Write control Programme – Scratch **S4A**

- Code is very similar to Arduino
- Block-based, so less likely to make mistakes



Scratch for Arduino



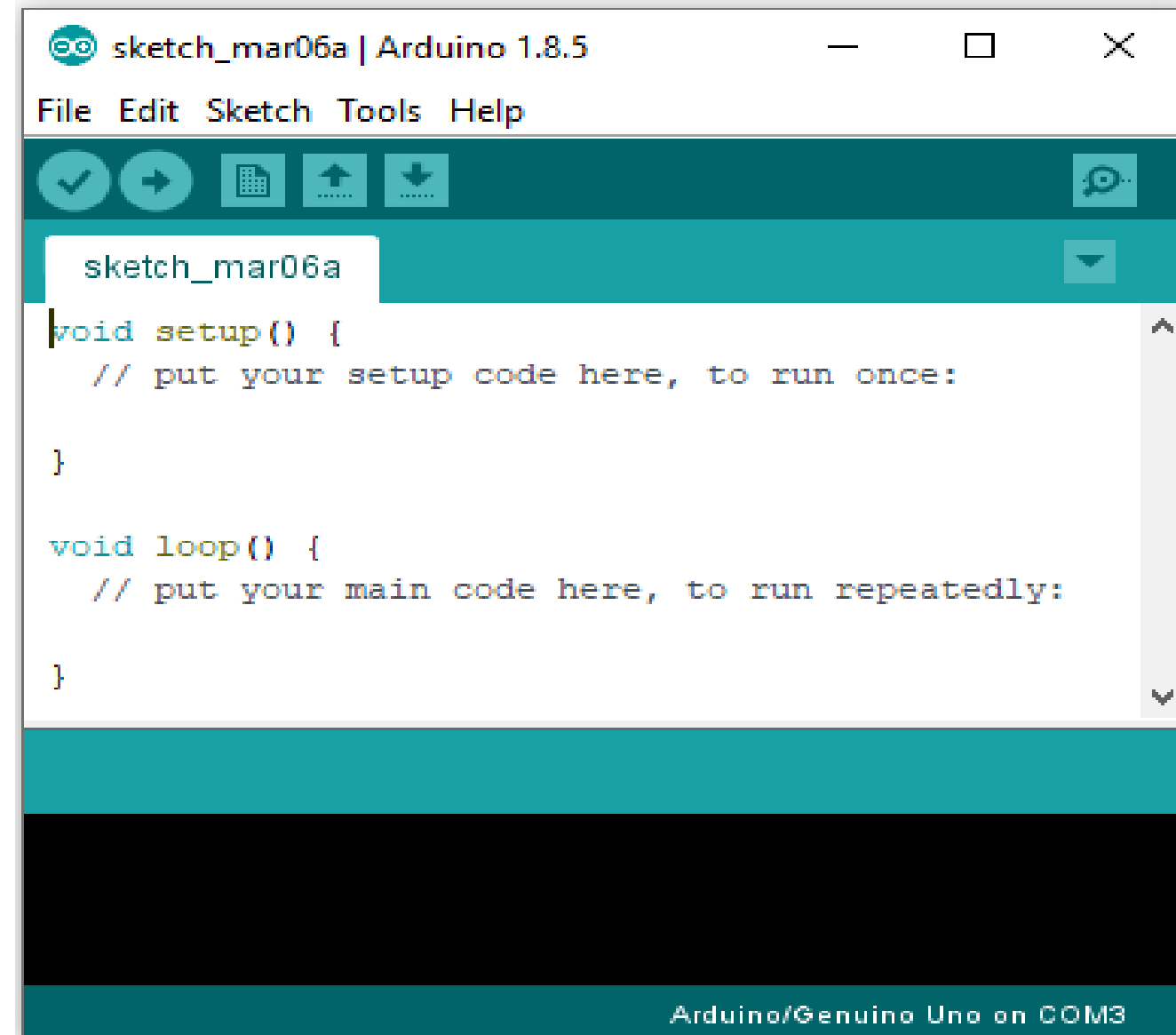
S4A



Programming Sketches - Cording

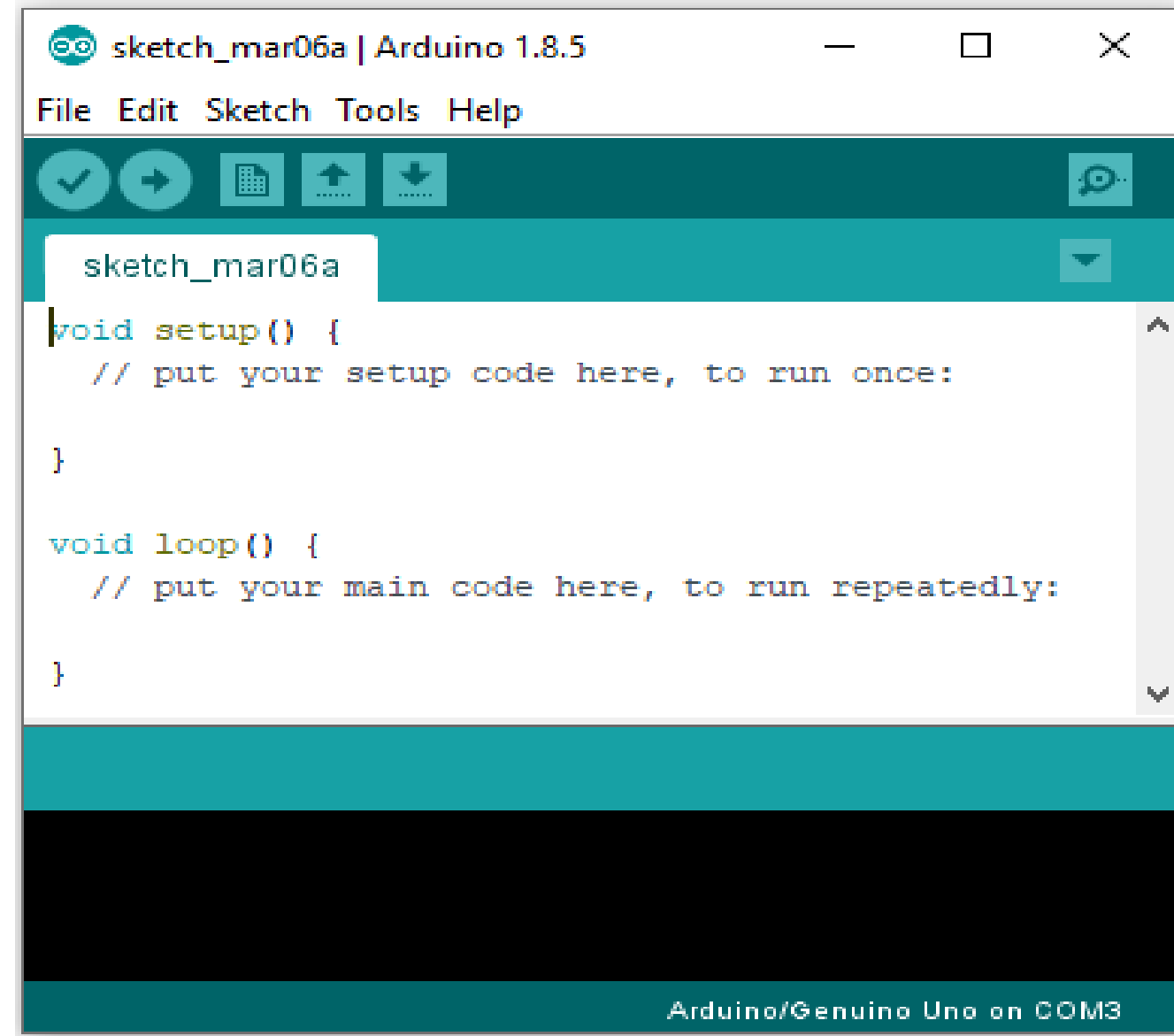
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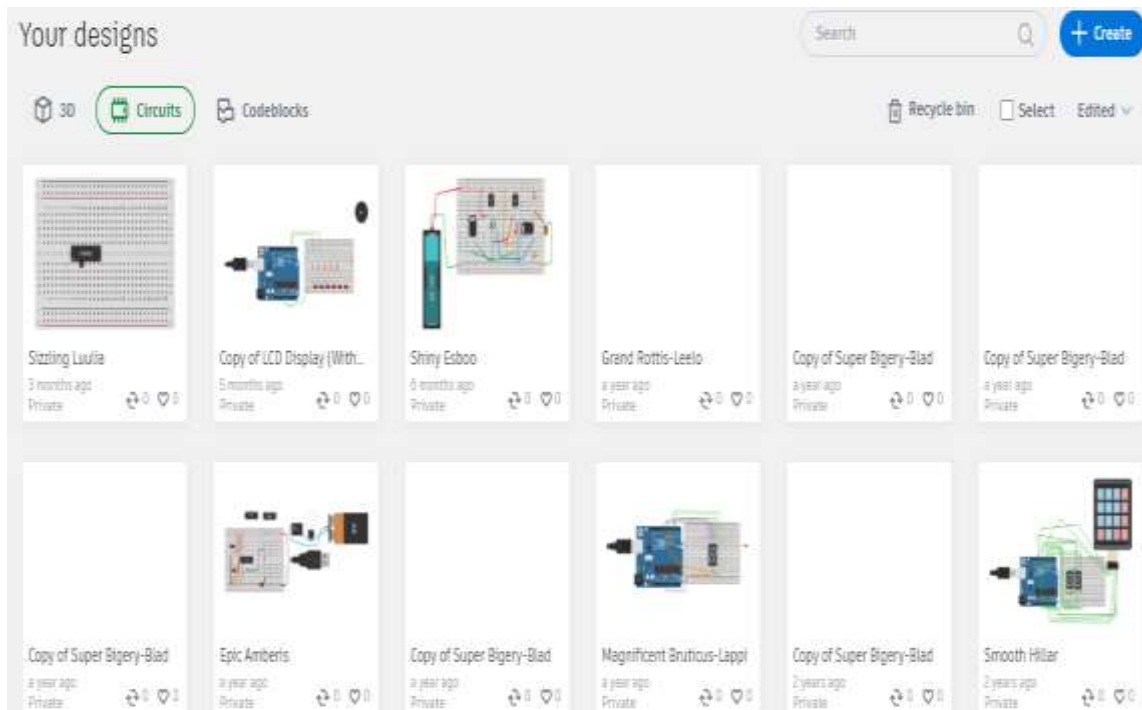
Programming Sketches - Cording

- Download & install the Arduino Software (IDE)
 - * <https://www.arduino.cc/en/software>

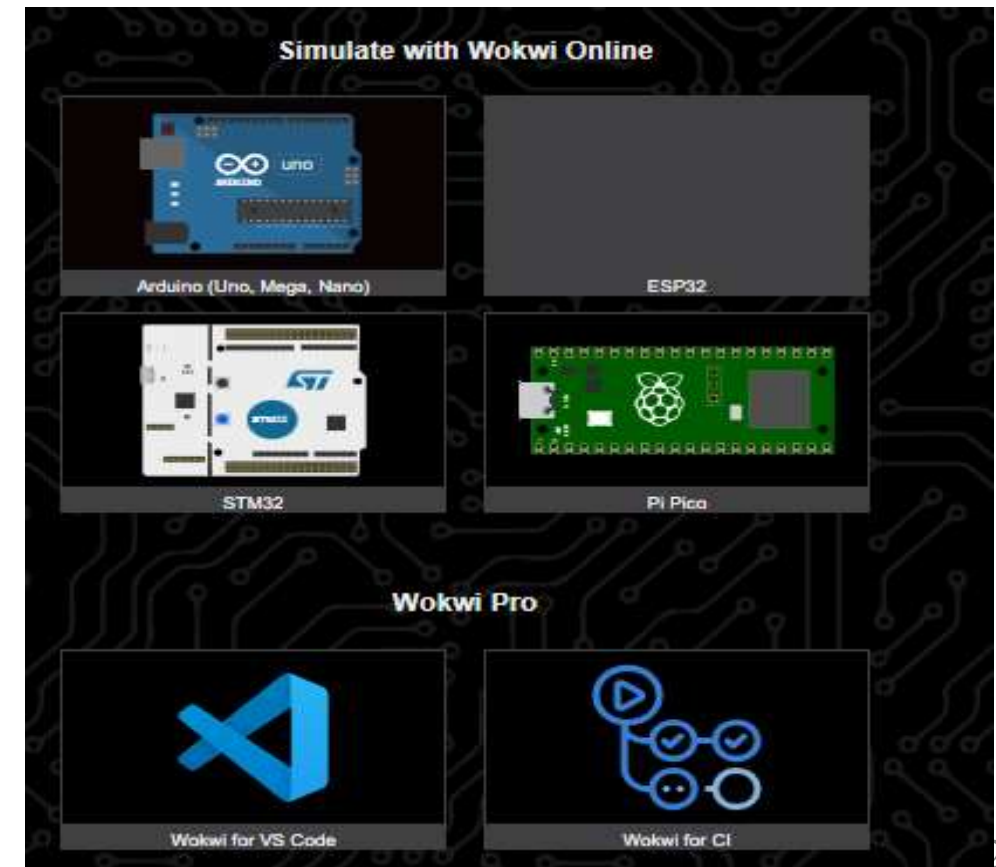


Online Tools:

<https://tinkercad.com>

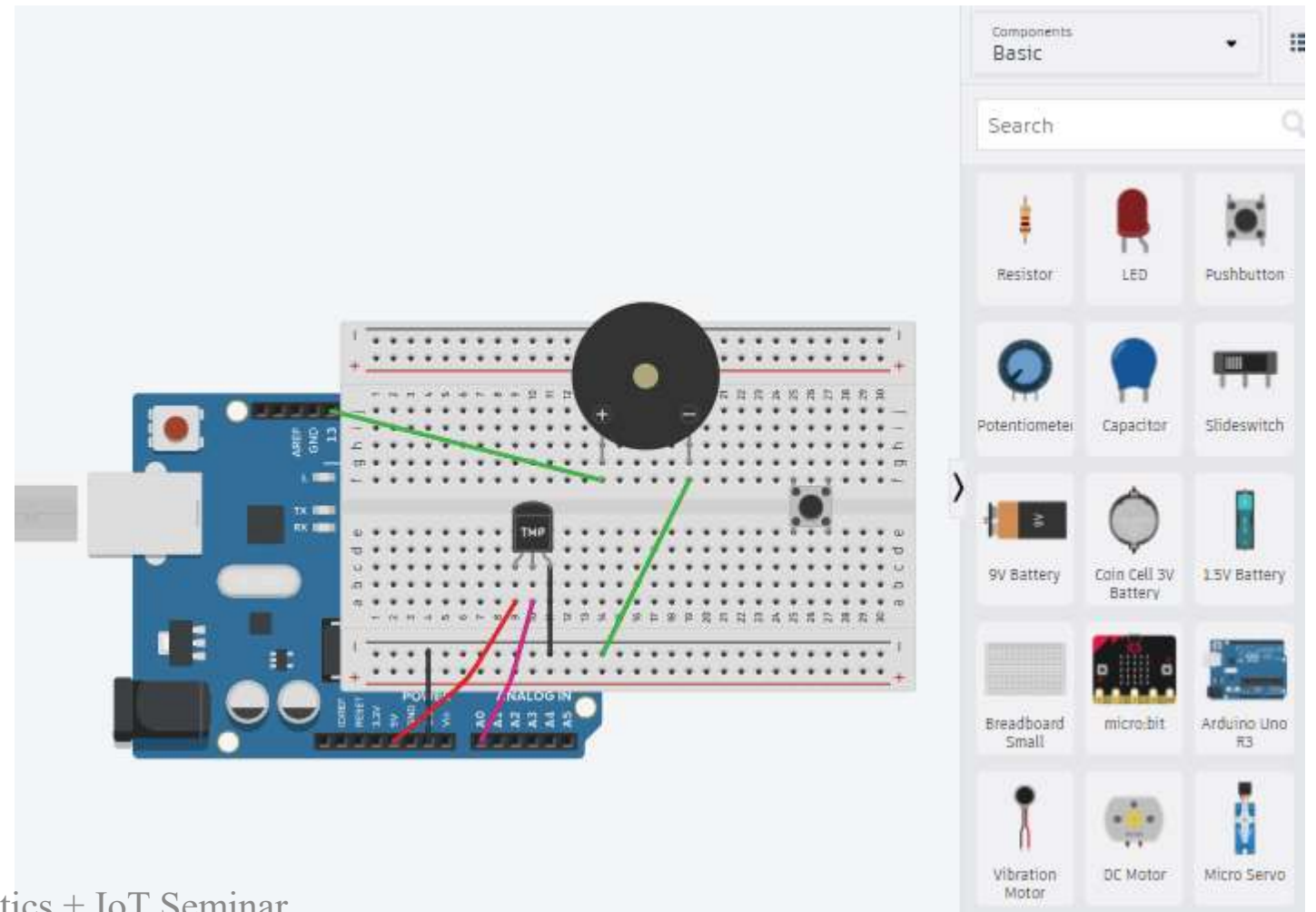
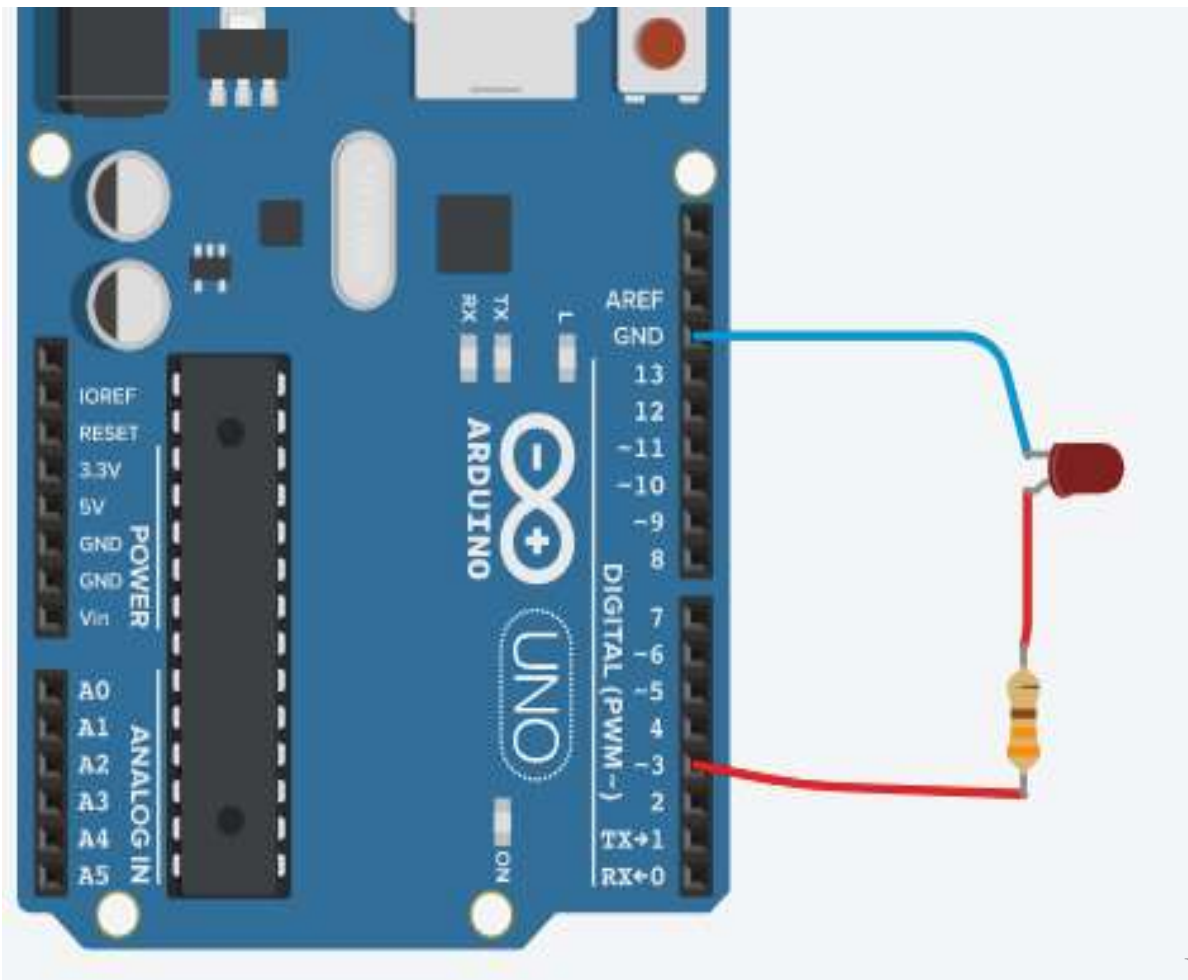


<https://wokwi.com/>



Sketch simulation on tinkercad

Use online simulation platform: <https://www.tinkercad.com/>



- Code runs in the computer and sends commands to the Arduino



Runs on Computer

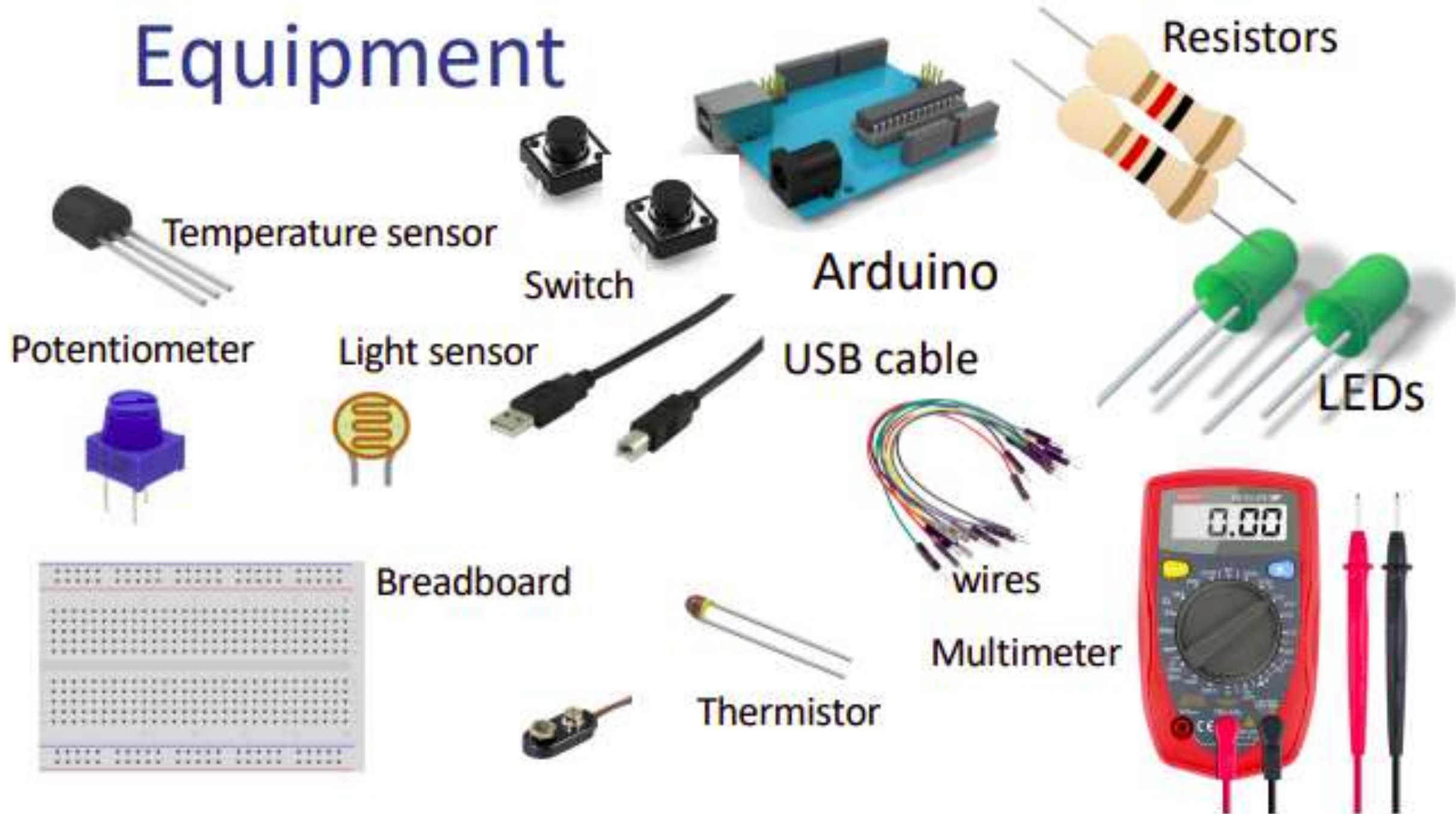
Commands
(eg. digital write)



Readings
(eg. digital read)

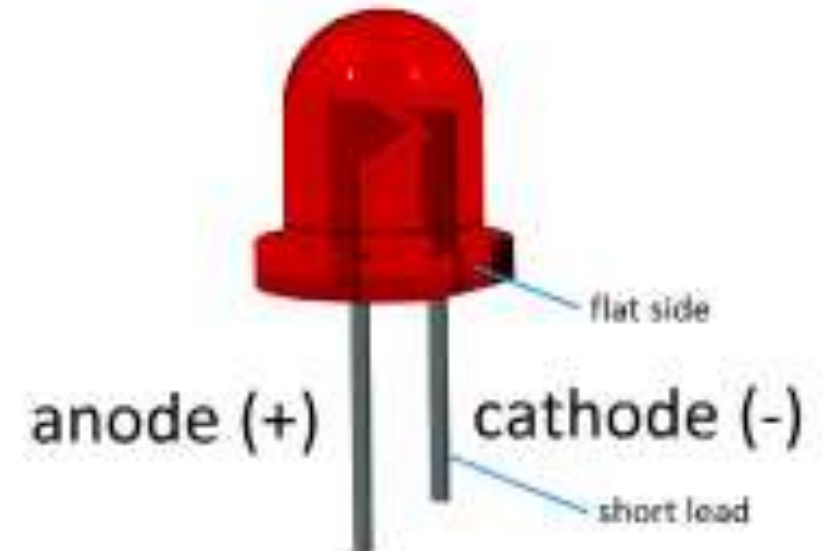
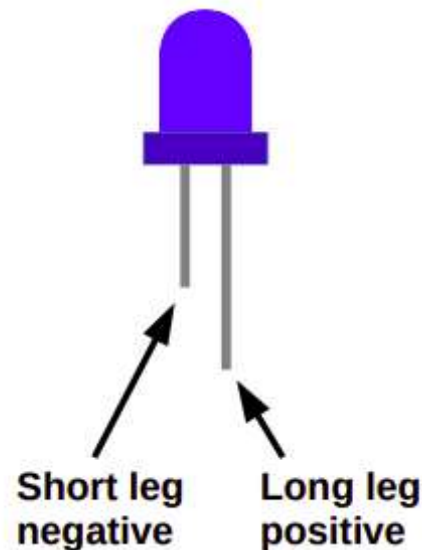


Equipment



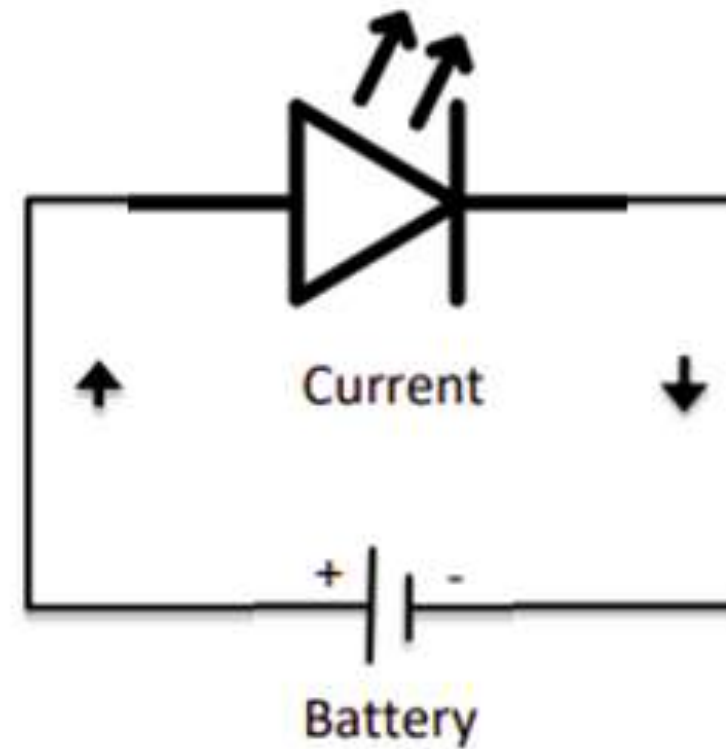
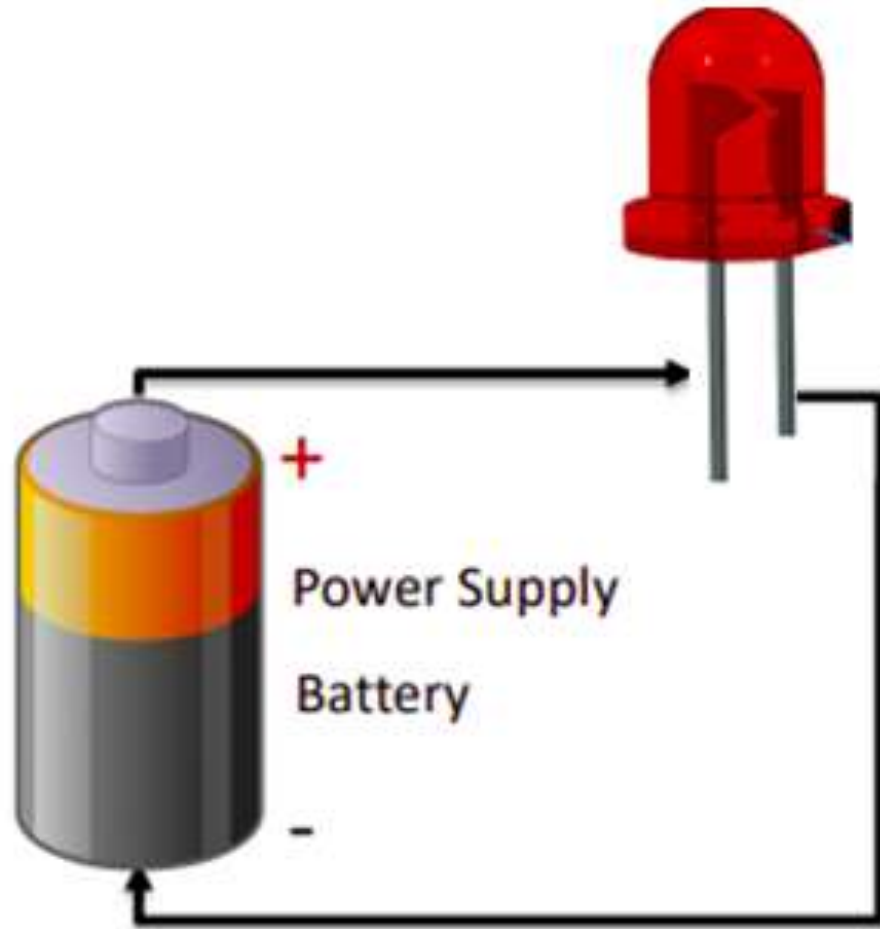
LED

- Light Emitting Diode
- Longer leg connects to positive
- Shorter leg connects to ground (0V)
- **Doesn't work if connected in reverse**

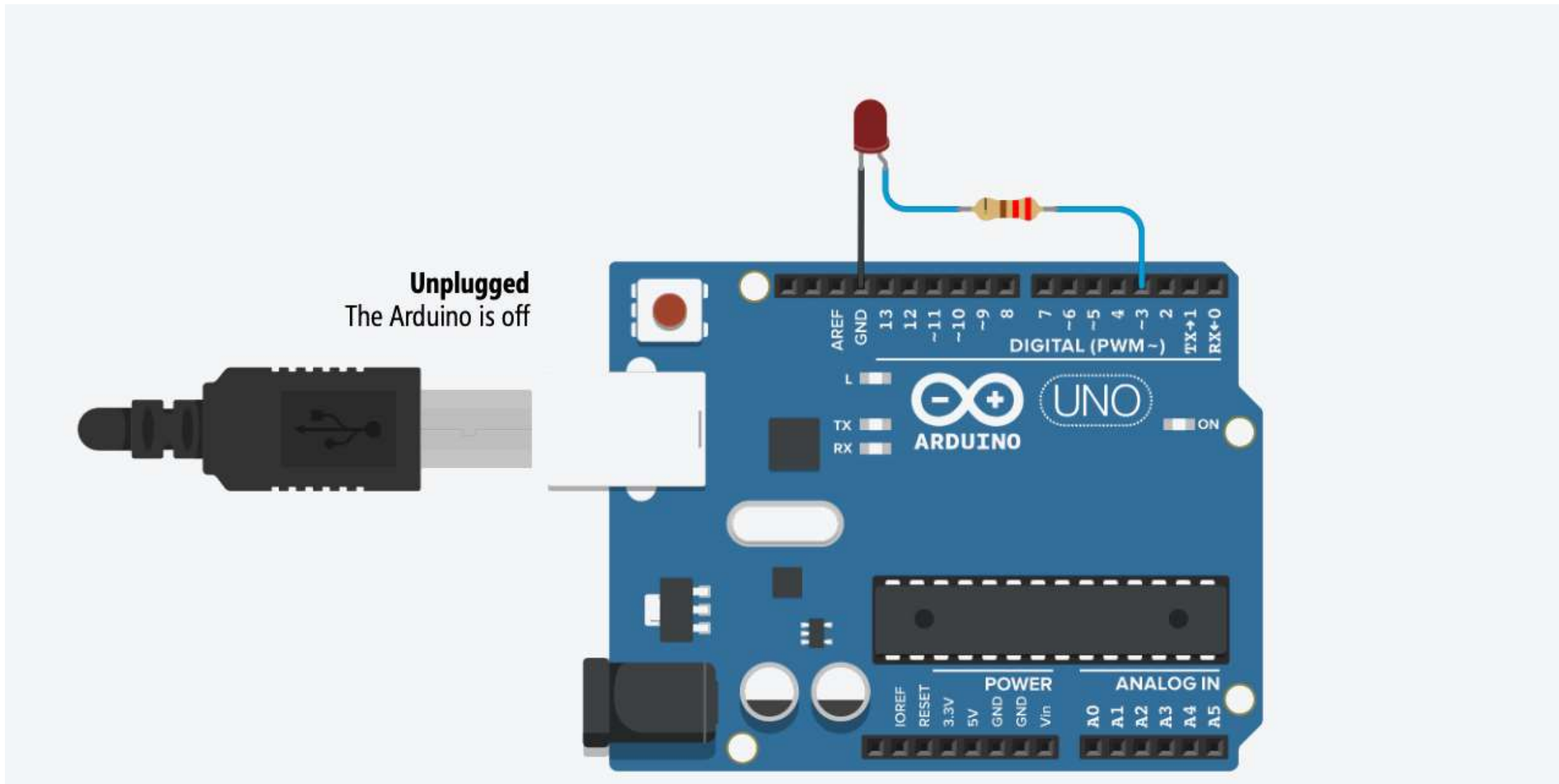


LED circuit

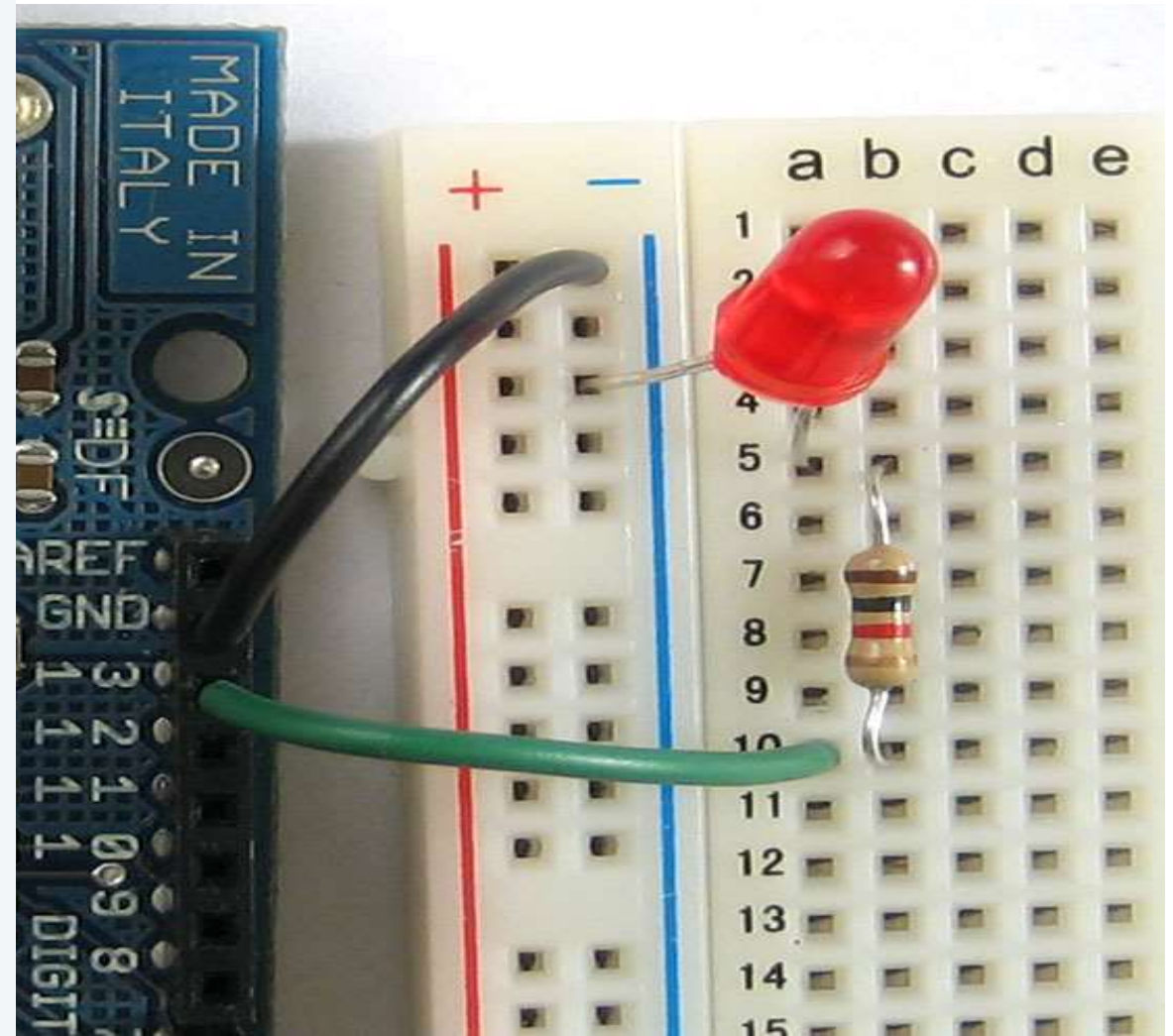
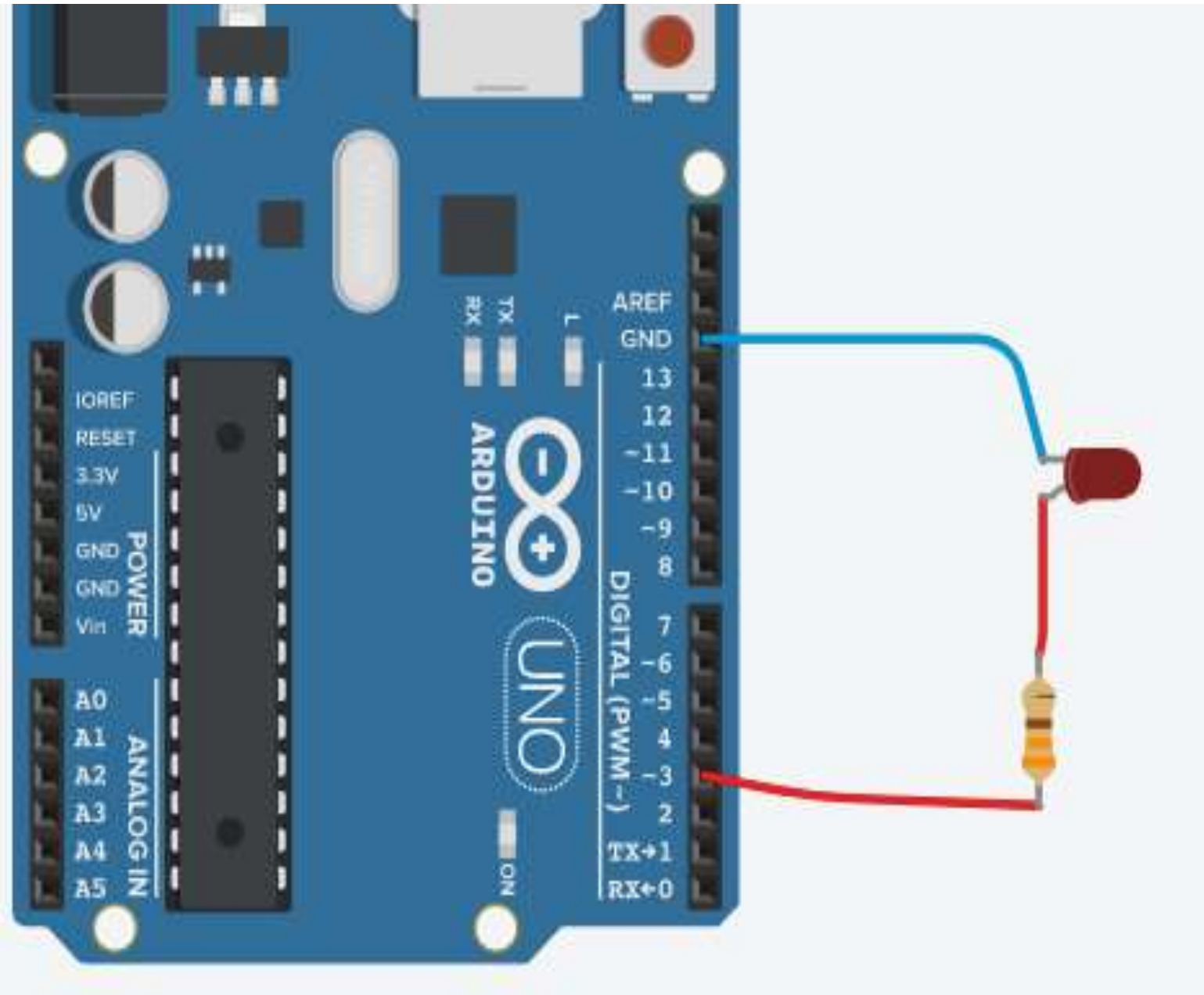
Here you see a basic Electrical Circuit:



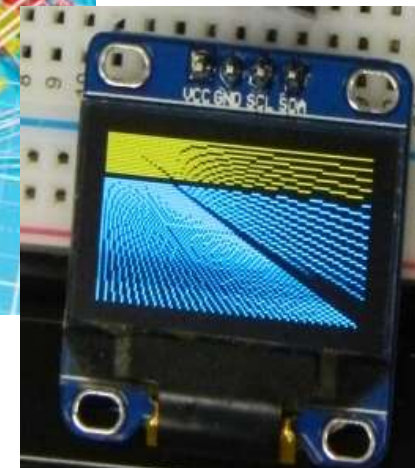
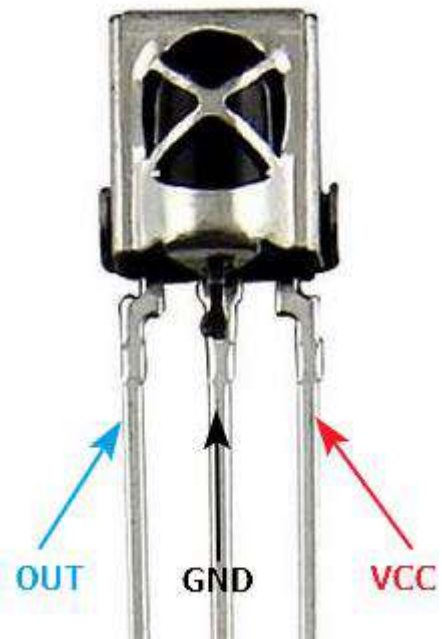
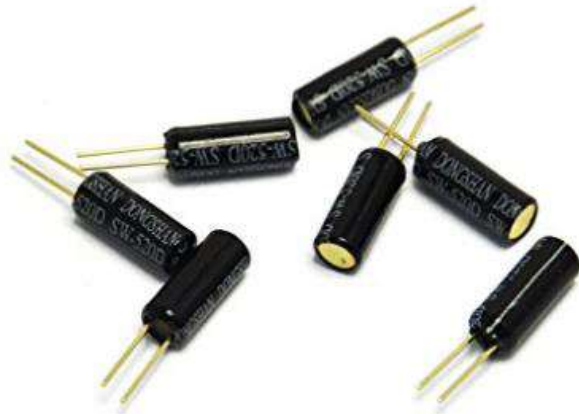
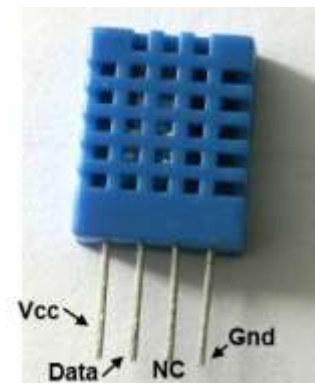
LED blinking circuit



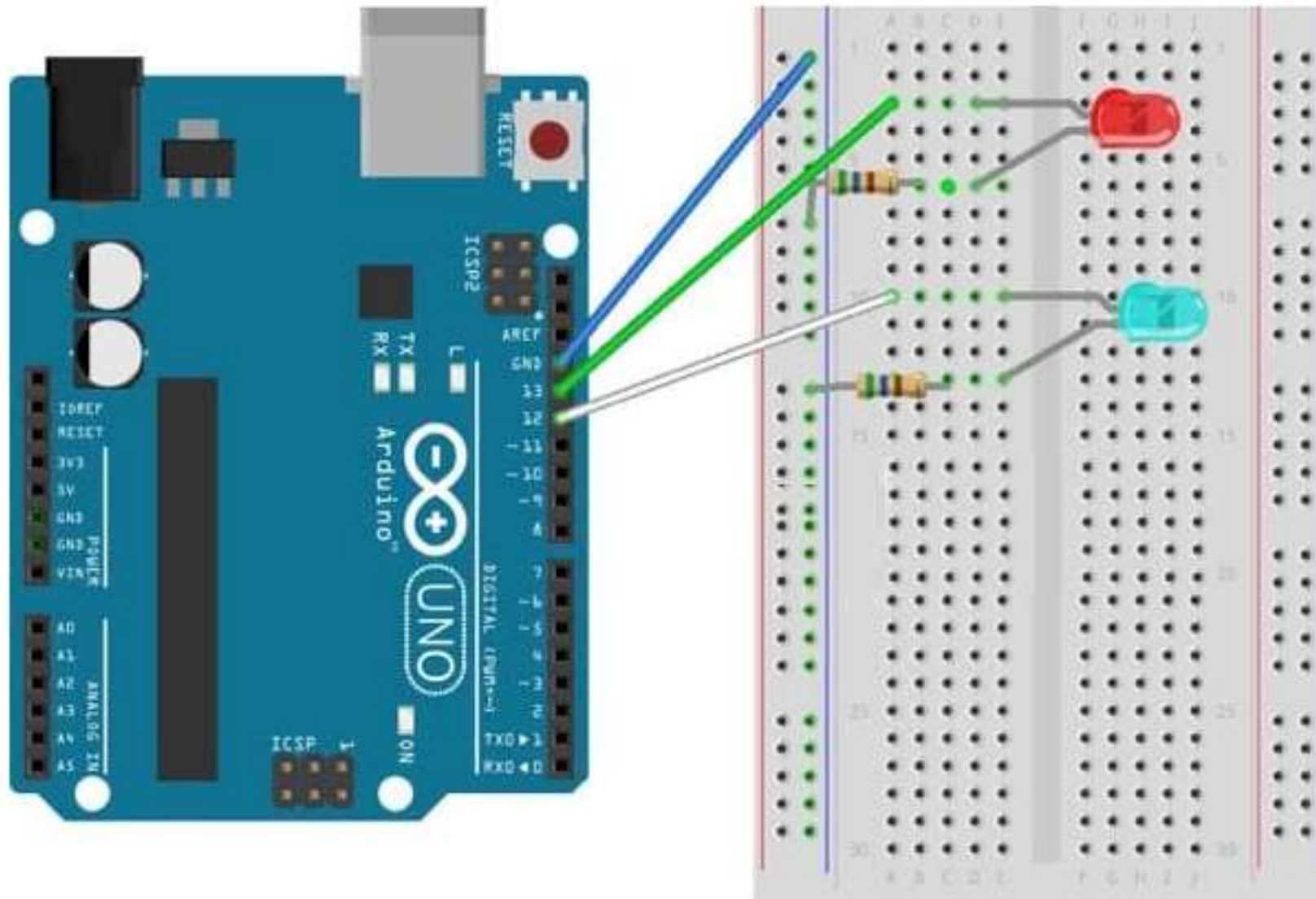
LED circuit on Bread Board



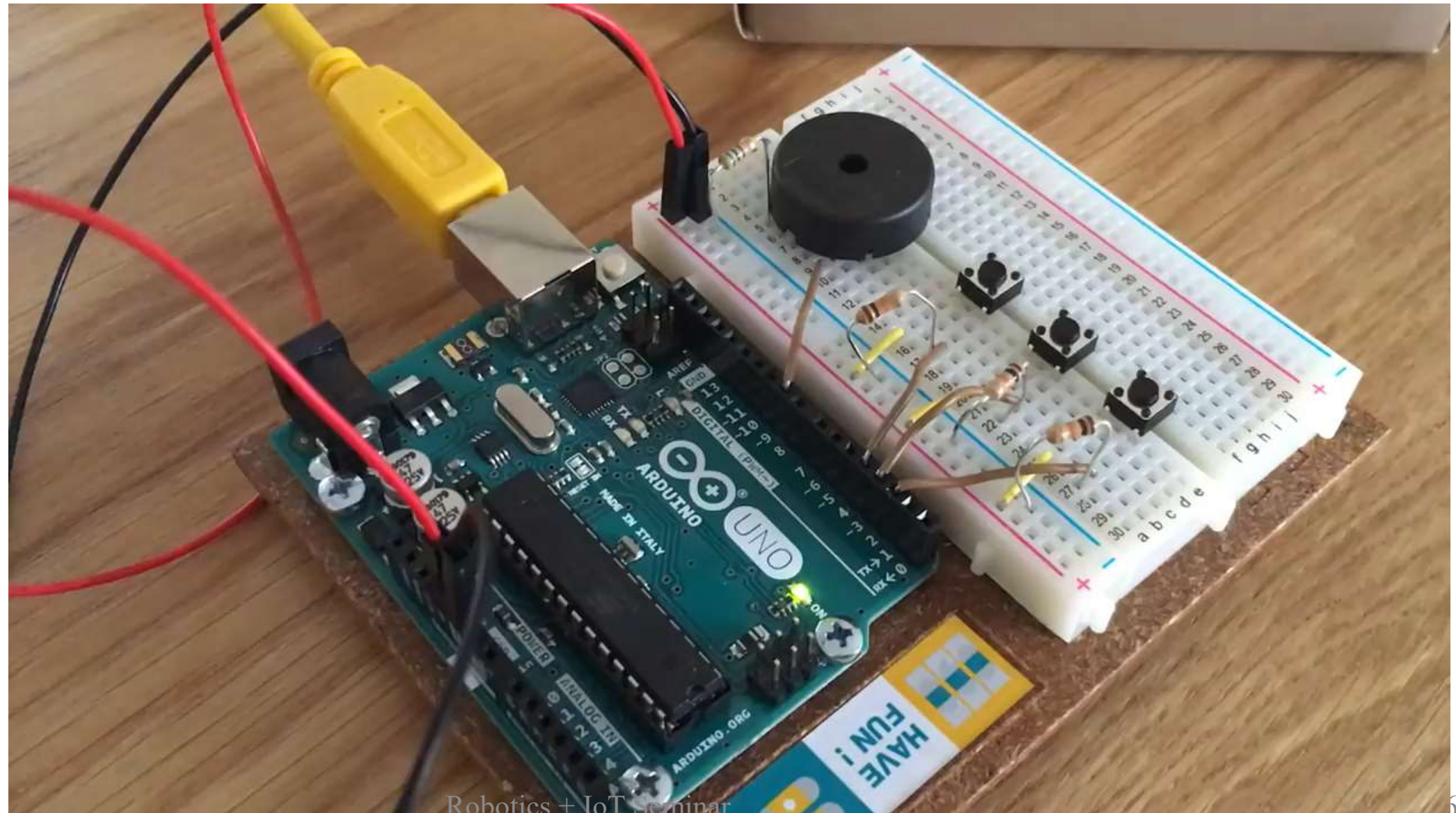
sensors and actuators

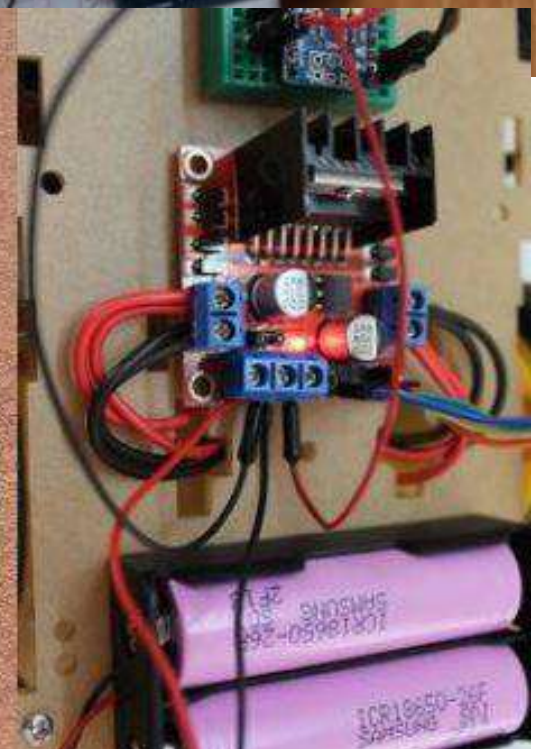
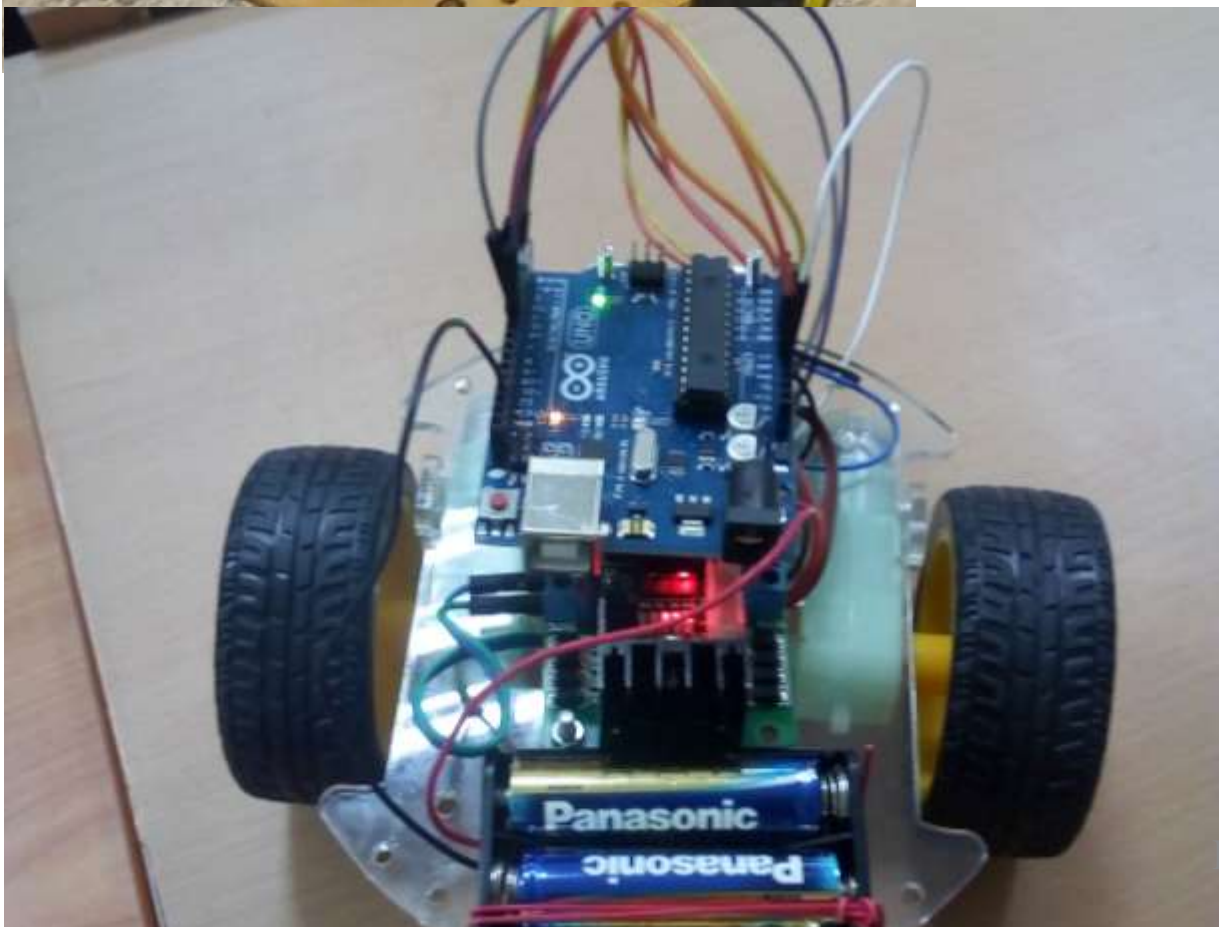
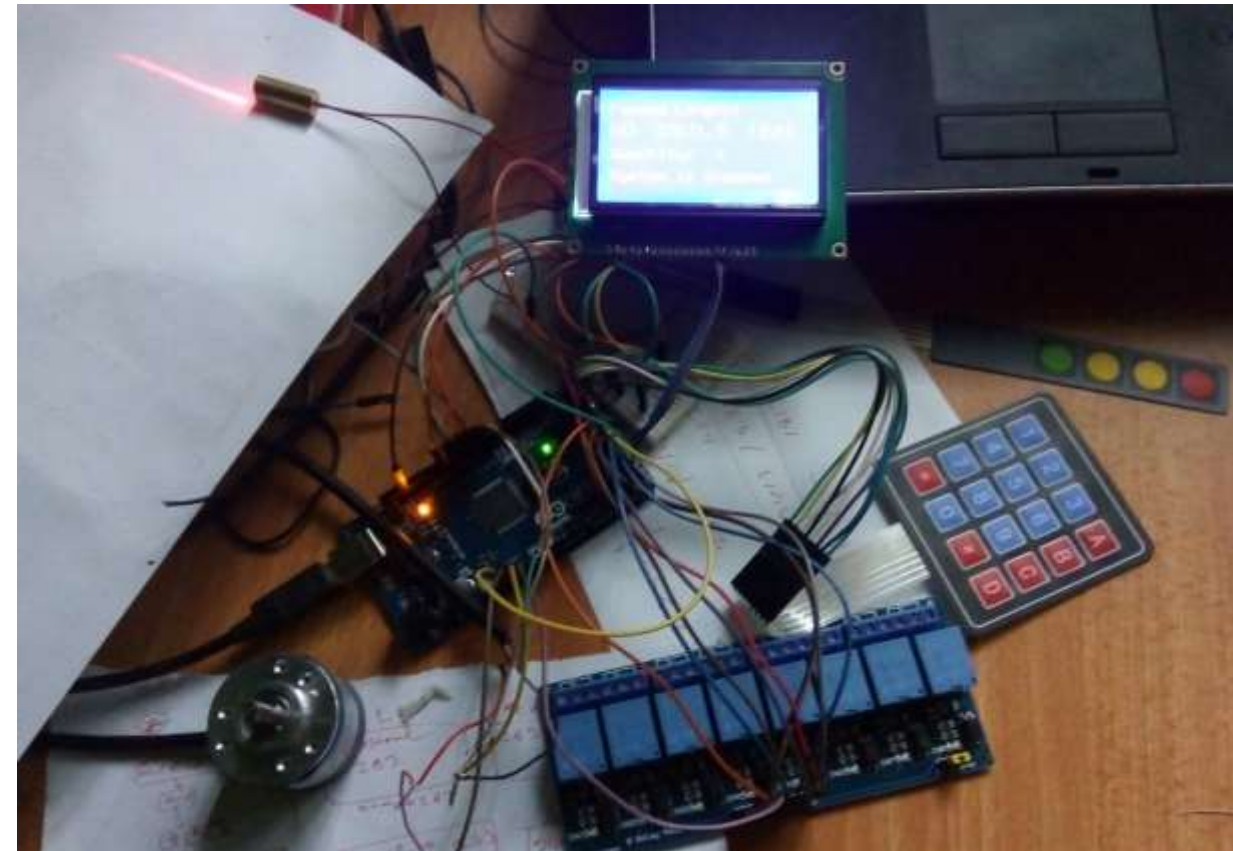


Simple LED circuit simulation



Simple Music circuit simulation







Contact Me:

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